

संघ २०८१

लोक सेवा आयोग

नेपाल इञ्जिनियरिङ्ग सेवा, सिभिल समूह, जनरल/हाइवे/स्यानिटरी/इरिगेशन/हाइड्रोपावर उपसमूह
राजपत्राङ्कित तृतीय श्रेणी (प्राविधिक) पदको प्रतियोगितात्मक लिखित परीक्षा

बिषय: Technical Subject | पत्र: द्वितीय | पूर्णाङ्क: १०० | समय: ३ घण्टा | मिति: २०८१/१०/१९

Section-A (30 Marks)

1. Define the permeability of soil and explain with neat sketches and formula. Elucidate the methods of determining the coefficient of permeability. 5

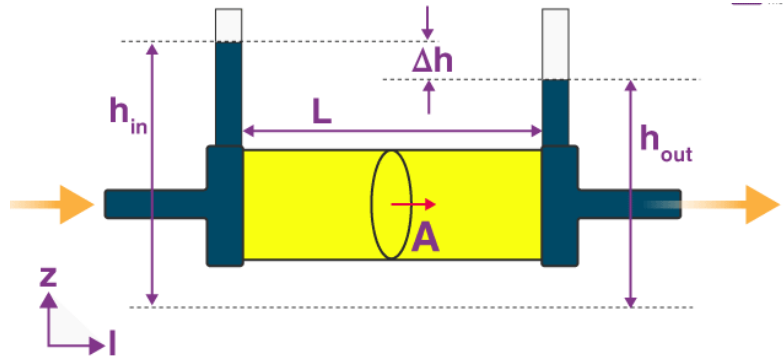
Answer:

A. Definition:

Permeability is defined as the property of a soil which permits the flow of water (or other fluids) through its interconnecting voids. A soil is said to be highly permeable if water passes through it quickly (e.g., gravel) and impermeable if the flow is extremely slow (e.g., clay).

B. Governing Law (Darcy's Law):

For laminar flow in saturated soils, the velocity of flow (v) is directly proportional to the hydraulic gradient (i):



Darcy's Law diagram

$$v = k \cdot i$$

Where:

- k = Coefficient of permeability (m/sec or cm/sec)
- i = Hydraulic gradient (h/L)
- Discharge $Q = k \cdot i \cdot A$

C. Methods of Determining Coefficient of Permeability (k):

- Laboratory Methods:
 - a) Constant Head Permeability Test: Suitable for coarse-grained soils (Gravel/Sand). $k = \frac{QL}{Aht}$.
 - b) Falling Head Permeability Test: Suitable for fine-grained soils (Silt/Clay). $k = 2.303 \frac{aL}{At} \log_{10} \left(\frac{h_1}{h_2} \right)$.
- Field Methods:
 - a) Pumping-out tests: (Using Dupuit's theory) for large-scale aquifer properties.
 - b) Pumping-in tests: (Packer tests) often used in dam site investigations.
- Indirect Methods: Calculation from grain size (Hazens Formula: $k = C \cdot D_{10}^2$) or consolidation test data.

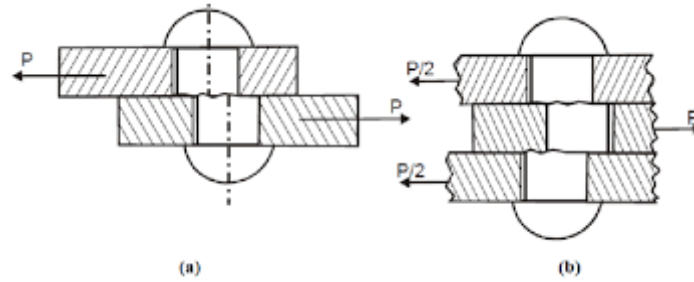
2. What are the three strengths of a riveted lap joint system of steel? How is the design strength of the joint determined? Explain with neat sketches and formula. 5

Answer:

In a riveted lap joint, the plates overlap and are connected by rivets. The strength of the joint is the minimum value of the following three failure modes:

a. Shearing Strength of Rivet (P_s):

The rivet fails by shearing across the interface of the plates. For a lap joint (single shear):

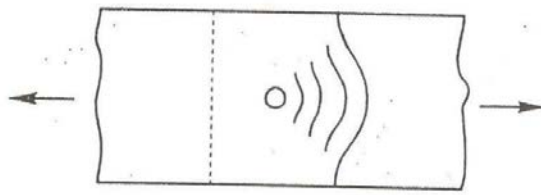


$$P_s = \frac{\pi d^2}{4} \cdot f_s$$

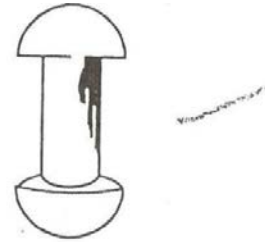
Where d = gross diameter of rivet, f_s = permissible shear stress.

b. Bearing Strength of Rivet/Plate (P_b):

The rivet or the plate crushes due to high contact pressure.



(e) Bearing Failure of Plate



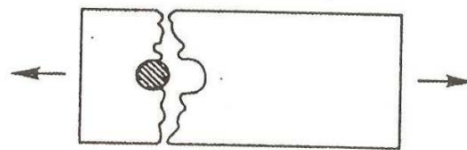
(f) Bearing Failure of Rivet

$$P_b = d \cdot t \cdot f_b$$

Where t = thickness of the thinner plate, f_b = permissible bearing stress.

c. Tearing Strength of Plate (P_t):

The plate tears across the row of rivets at the net section.



(c) Tearing Failure of Plate

$$P_t = (p - d) \cdot t \cdot f_t$$

Where p = pitch of rivets, f_t = permissible tensile stress.

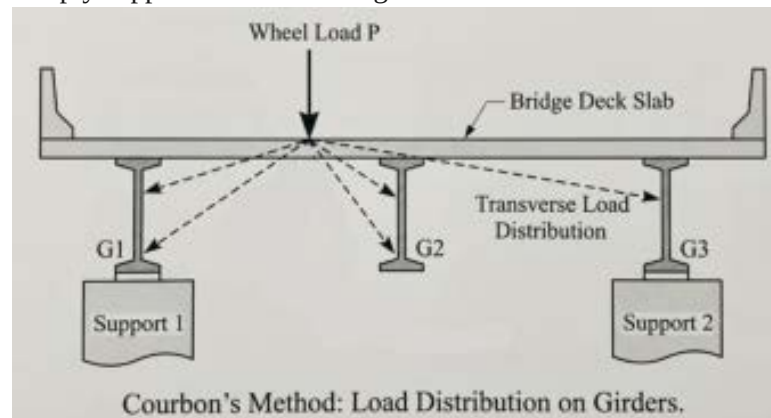
Determination of Design Strength:

- The Design Strength of the Joint is the least of P_s , P_b , and P_t .
- Efficiency (η): $\eta = \frac{\text{Least of } P_s, P_b, P_t}{\text{Strength of solid plate}(p \cdot t \cdot f_t)} \times 100\%$.

3. Mention the major principles and steps to be followed in the design of T-beam bridge using carbon's method. 10

Answer:

Courbon's Method is a classical, popular manual approach for analyzing the lateral load distribution among longitudinal girders in simply supported T-beam bridges.



a. MAJOR PRINCIPLES & ASSUMPTIONS

The method is based on the following mechanical principles:

- **Transverse Rigidity:** The bridge deck is assumed to be infinitely stiff in the transverse direction.
- **Linear Deflection Profile:** The transverse profile of the bridge deck remains straight after the application of load.
- **Integrated Action:** Longitudinal girders are assumed to be connected by infinitely stiff cross-beams or diaphragms.
- **Load Sharing:** Due to the rigid rotation/translation of the deck, loads are shared among girders based on their relative flexural stiffness and their distance from the bridge's central axis.

b. CONDITIONS FOR VALIDITY

Courbon's Method is applicable only when the following criteria are satisfied:

- a) **Aspect Ratio (L/B):** The ratio of span (L) to the width of the deck (B) must be greater than 2 but less than 4 ($2 < L/B < 4$).
- b) **Cross-Girders:** The longitudinal girders must be interconnected by at least **five** symmetrically spaced cross-girders.
- c) **Stiffness of Cross-Girders:** The depth of the cross-girders must be at least **0.75 times** the depth of the main longitudinal girders.

c. REACTION CALCULATION FORMULA

The reaction shared by a specific longitudinal girder (R_i) is calculated as:

$$R_i = \frac{\sum W}{n} \left[1 + \frac{n \cdot I \cdot e \cdot d_i}{\sum I \cdot d_i^2} \right]$$

Where:

- $\sum W$: Total concentrated live load.
- n : Number of longitudinal girders.
- I : Moment of inertia of each longitudinal girder.
- e : Eccentricity of live load with respect to the bridge axis.
- d_x or h : Distance of the girder under consideration from the central axis.

DESIGN STEPS FOR T-BEAM BRIDGE

The design follows a sequential process from preliminary sizing to detailed reinforcement:

Step 1: Planning and Preliminary Sizing

- **Main Girder:** Depth (D) is usually taken as 1/10 to 1/15 of the span for simple spans. Web width (b_w) should be ≥ 250 mm.
- **Cross Girder:** Depth $\geq 0.75D$; spacing between 3m to 5m.
- **Deck Slab:** Minimum thickness of 200 mm.
- **Effective Span:** Clear span + bearing width or clear span + effective depth, whichever is less.

Step 2: Analysis and Design of Deck Slab

- **Cantilever Slab:** Analyzed using the **Effective Width Method**.
- **Restrained Slab:** Analyzed using **Pigeaud's Method** (assuming support on four sides).
- Design for Bending Moment (M_u) and check for Shear.

Step 3: Analysis of Main Girders

- **Dead Load:** Calculate self-weight of slab, girders, wearing course, and railings.
- **Live Load Distribution:** Apply **Courbon's Method** to determine the reaction factor and share of live load for each girder.
- **Critical Position:** Position the IRC train loads (Class AA, 70R, or Class A) longitudinally and transversely to produce maximum M_u and V_u .

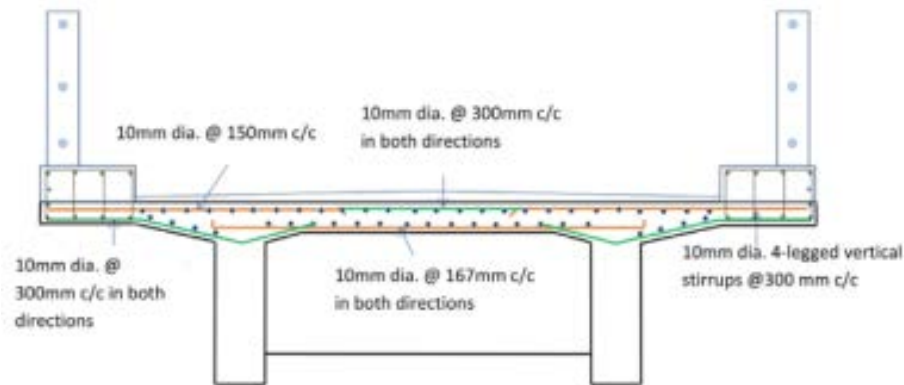
Step 4: Design of Main Girders

- **Bending:** Design as a flanged (T) section. Determine the Neutral Axis (x_u); if $x_u < D_f$, design as a rectangular section of width b_f .
- **Shear:** Design vertical stirrups for the net shear force. Check that the nominal shear stress (τ_v) does not exceed $\tau_{c,max}$.
- **Serviceability:** Perform mandatory checks for **Deflection** and **Crack Width**.

Step 5: Design of Cross Girders

- Analyze for dead loads and the share of live loads transmitted from the slab.
- Design for flexure and shear at support and mid-span.

Step 6: Detailing



- Provide longitudinal reinforcement with proper anchorage (L_d) at supports.
- Provide side face reinforcement if the depth exceeds 750 mm.
- Detail the Curtailment of bars based on the bending moment envelope.

4. A site investigation has to be done for a 6-lane bridge to be constructed in Karnali river. The client seeks the information on bearing capacity and settlement of foundation for the bridge. What method of site investigation and in-situ tests would you advise as a project engineer? Explain the in-situ test you advised in detail. 10

Answer:

a. Recommended Site Investigation Program:

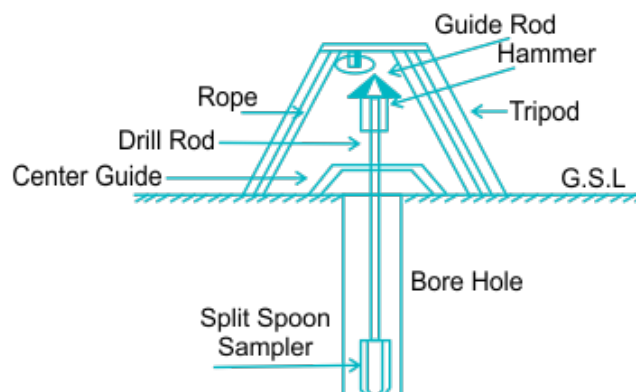
Due to the high discharge, potential for deep scour, and boulders typical of the Karnali river, the following are advised:

- **Boring:** Rotary Core Drilling (to reach bedrock or firm strata).
- **Geophysical Survey:** Seismic Refraction to map the bedrock profile across the wide river bed.
- **In-situ Tests:** **Standard Penetration Test (SPT)** for sandy/silty layers and **Dynamic Cone Penetration Test (DCPT)** if gravelly. Given the scale, a **Plate Load Test** on the bank or cofferdam area is advised for settlement data.

b. Detailed Explanation of Advised In-situ Test: Standard Penetration Test (SPT)

The SPT is the most widely used in-situ test to determine the bearing capacity and settlement of cohesionless soils.

- **Apparatus:** Split spoon sampler, Drop hammer (63.5 kg), Guiding rod, and Drilling rig.



- **Procedure:**
 - a) A borehole is drilled to the desired depth.
 - b) The sampler is driven into the soil by the 63.5 kg hammer falling from a height of **750 mm**.
 - c) The number of blows required for three 150 mm penetrations are recorded.
 - d) The **N-value** is the sum of blows for the last 300 mm (ignoring the first 150 mm as seating drive).
- **Corrections Applied:**
 - a) **Overburden Pressure Correction:** $N_c = N \cdot C_n$.

b) **Dilatancy Correction:** (For fine sand/silt below water table) $N' = 15 + \frac{1}{2}(N_c - 15)$.

• **Application:**

- o **Bearing Capacity:** Correlated with angle of internal friction (ϕ) and N-values.
- o **Settlement:** Empirical formulas relate N-values to the settlement of footings.

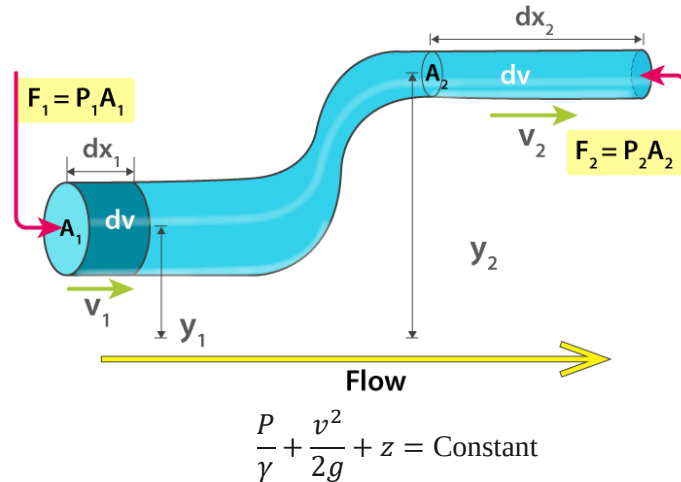
Section-B (25 Marks)

5. Explain the concept of Bernoulli's equation. Show each component of Bernoulli's equation in case of a uniform diameter of pipe flow. Also, draw the HGL and TEL for this case. 5

Answer:

a. Concept:

Bernoulli's Equation states that for an incompressible, non-viscous, irrotational, and steady flow, the sum of pressure energy, kinetic energy, and potential energy per unit weight is constant along a streamline.



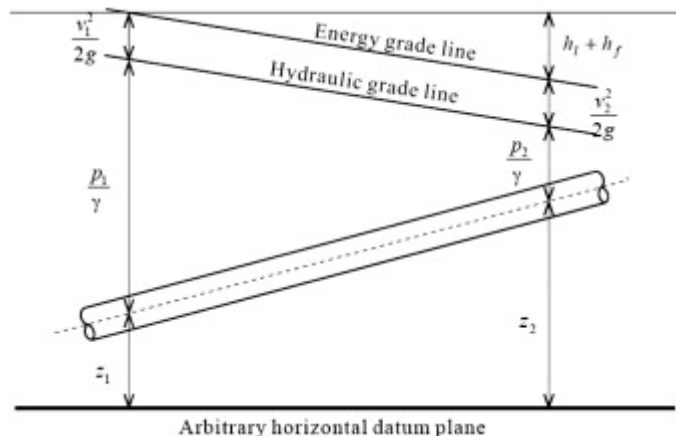
b. Components:

- Pressure Head ($\frac{P}{\gamma}$): Energy due to static pressure.
- Velocity Head ($\frac{v^2}{2g}$): Energy due to motion.
- Datum Head (z): Potential energy due to elevation.

c. Case of Uniform Diameter Pipe:

In a uniform pipe, by continuity ($A_1 v_1 = A_2 v_2$), the velocity (v) is constant. Thus, the Velocity Head remains constant throughout the length.

d. HGL and TEL:



- Total Energy Line (TEL): Sum of all three heads. It always slopes downwards in the direction of flow due to friction loss (h_f).
- Hydraulic Gradient Line (HGL): Sum of Pressure Head and Datum Head. In a uniform pipe, HGL is parallel to TEL, separated by a constant distance of $\frac{v^2}{2g}$.

6. Describe the significance of hydrological analysis for a hydropower project. What are the different parameters of hydrological analysis? How do you estimate and use these parameters in the project? State. 3+3+4=10

Answer:

- Significance:
 - a) Energy Estimation: It estimates the magnitude and variability of river flows, which directly determines the power potential ($P = \eta \rho g Q H$).
 - b) Safety: Determines the Design Flood for spillway sizing.
 - c) Optimization: Helps in sizing the reservoir, intake, and settling basin.
 - d) Risk Management: It identifies extreme weather patterns, such as floods and droughts, ensuring structures are robust enough to withstand high flows and resilient during dry periods.
 - e) Regulatory & Environmental Compliance: Analysis ensures projects meet legal requirements for water usage and maintain necessary environmental flows to support downstream ecosystems.
- Parameters and Estimation:

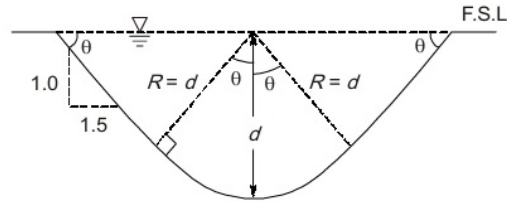
Parameter	Estimation Method	Use in Project
Streamflow (River Discharge)	Gauging stations, current meters, or area-velocity method.	Basic data for calculating power ($P = \eta \rho g Q H$) and seasonal energy distribution.
Precipitation	Rain gauges (manual/automatic) and satellite-based data (TRMM/GPM).	Used to predict runoff patterns, seasonal water availability, and fill gaps in missing flow data.
Catchment Area	Topographic maps, Digital Elevation Models (DEM), and GIS tools.	Determines the total watershed size to estimate total water yield and runoff potential.
Design Discharge (Q_d)	Flow Duration Curve (FDC) analysis (e.g., Q_{40} or Q_{45} percentage).	Sizing of turbines, generators, and penstock; determines the plant's installed capacity.
Design Flood (Q_f)	Gumbel's distribution, Flood Frequency Analysis (100/1000 year), or PMF.	Critical for designing spillways, diversion tunnels, and determining dam height.
Flow Duration Curve (FDC)	Sorting historical daily/monthly flow data in descending order of magnitude.	Graphical tool to decide the "firm power" and "secondary power" potential of the site.
Sedimentation Rate	Sediment rating curves and catchment erosion analysis.	Designing settling basins (desanders) and calculating reservoir life (dead storage).
Evaporation/Losses	Pan evaporimeters and empirical formulas (e.g., Penman-Monteith).	Calculating net water volume in reservoirs to account for atmospheric and seepage losses.
Environmental Flow (Q_{env})	Regulatory minimums (e.g., 10% of lean season flow) or habitat modeling.	Ensures ecological sustainability and legal compliance for water release.

- Application in the Project:
 - a) **Hydraulic Design:** Parameters like Streamflow and Design Discharge are used to calculate the diameter of the penstock and the specific speed of the turbines.
 - b) **Safety Engineering:** The Design Flood is used to ensure the spillway can handle peak discharge without overtopping the dam, preventing catastrophic failure.
 - c) **Economic Planning:** The Flow Duration Curve helps in predicting annual energy generation, which is used to calculate the Internal Rate of Return (IRR) and Benefit-Cost ratio.
 - d) **Maintenance Strategy:** The Sedimentation Rate determines the frequency of flushing operations and the design of turbine blades to resist silt erosion.

7. Design concrete-lined irrigation canal to carry a discharge of 45 cumes. The bed slope of canal is 1:3000. Take side slope 2:1 and Manning's coefficient $\eta = 0.015$. Assume any data you need. 10

Answer:

- a. Assumptions:



Area, $A = d^2(\theta + \cot \theta)$
 Perimeter, $P = 2d(\theta + \cot \theta)$

- Use a trapezoidal section (most economical or practical).
 - Assume a trial depth (d) or b/d ratio. Let's try $b/d = 4$.
- b. Geometric Properties:
- Side slope $m = 2$.
 - Area $A = (b + md)d = (4d + 2d)d = 6d^2$.
 - Wetted Perimeter $P = b + 2d\sqrt{1 + m^2} = 4d + 2d\sqrt{5} \approx 8.47d$.
 - Hydraulic Radius $R = A/P = 6d^2/8.47d \approx 0.708d$.
- c. Manning's Equation:

$$Q = \frac{1}{n} AR^{2/3} S^{1/2}$$

$$45 = \frac{1}{0.015} (6d^2)(0.708d)^{2/3} (1/3000)^{1/2}$$

$$45 = 66.67 \times 6d^2 \times 0.794d^{2/3} \times 0.01825$$

$$45 = 5.797d^{8/3} \Rightarrow d^{8/3} \approx 7.76 \Rightarrow d \approx 2.15 \text{ m}$$

- d. Dimensions:
- Depth $d = 2.15 \text{ m}$.
 - Width $b = 4 \times 2.15 = 8.6 \text{ m}$.
 - Velocity $v = Q/A = 45/(6 \times 2.15^2) \approx 1.62 \text{ m/s}$ (safe for concrete lining).
 - Add Freeboard $\approx 0.75 \text{ m}$.

Section-C (25 Marks)

8. Write down about California Bearing Ratio (CBR) test. How is a pavement designed with CBR value? 2+3=5

Answer:

a. CBR Test:

It is a penetration test developed to evaluate the subgrade strength of roads. A standard piston (50 mm dia) is penetrated into soil at 1.25 mm/min.

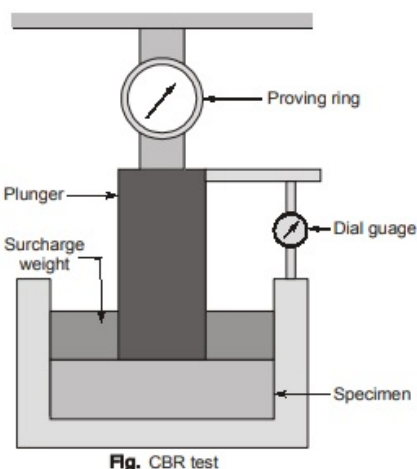
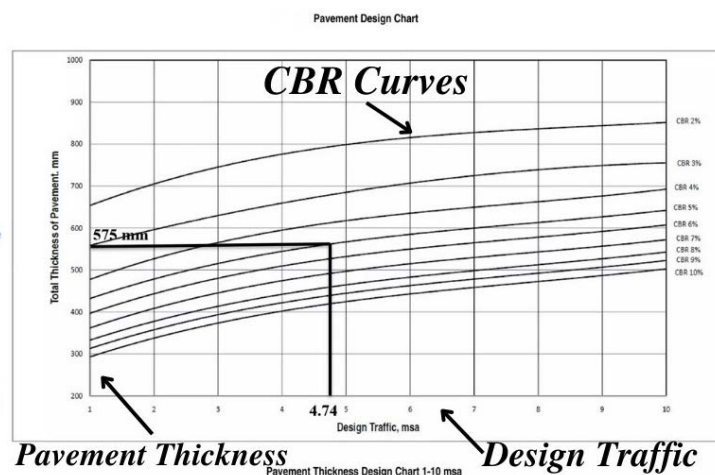


Fig. CBR test



$$\text{CBR}(\%) = \frac{\text{Pressure sustained by specimen at 2.5 or 5.0 mm penetration}}{\text{Standard pressure by crushed stones at same penetration}} \times 100$$

(Standard pressures: 70 kg/cm² for 2.5 mm; 105 kg/cm² for 5.0 mm).

b. Pavement Design:

- Principle: A higher CBR value indicates a stronger subgrade, requiring a thinner pavement.

- **Design Curves:** Standard IRC design charts (like IRC:37) are used where the x -axis is Cumulative Standard Axles (CSA) and the y -axis is the total thickness. Different curves represent different CBR values (2% to 10%+).

- **Formula (Old California method):** $T = \sqrt{\frac{1.75P}{CBR} - \frac{A}{\pi}}$ (where P is wheel load).

9. State traffic control devices that are used on roads with their functions. Factors to be considered while designing a road intersection? Elaborate. 4+6=10

Answer:

I. Traffic Control Devices and Their Functions

Traffic control devices are mediums used to communicate with road users to ensure the safe and efficient movement of traffic.

- **Traffic Signs:** Categorized into three types:
 - o **Regulatory Signs:** Mandatory signs (e.g., Stop, No Entry) that carry legal requirements; violations are punishable.
 - o **Warning/Cautionary Signs:** Alert drivers to hazardous conditions ahead (e.g., Narrow Bridge, Sharp Curve).
 - o **Informative Signs:** Provide guidance regarding destinations, distances, and services (e.g., Hospital, Petrol Pump).
- **Traffic Signals:** Control the right-of-way by allocating time intervals to different traffic streams, reducing right-angle collisions at high-volume intersections.
- **Road Markings:** Applied to the pavement to provide continuous guidance. They define lanes, indicate overtaking zones (solid vs. broken lines), and mark pedestrian paths (Zebra crossings).
- **Traffic Islands/Channelizers:** Physical structures used to segregate traffic into specific paths, reducing the "conflict area" and controlling the angle of merging and diverging.

II. Factors Considered in Road Intersection Design

The primary objective of intersection design is to reduce "conflict points" (locations where vehicle paths cross) while maintaining traffic flow.

- **Traffic Volume and Composition:** Design must be based on Peak Hour Traffic. Traffic is converted into **Passenger Car Units (PCU)** to account for the varying sizes of vehicles (trucks vs. motorcycles).
- **Conflict Point Management:** A standard four-legged intersection has **32 conflict points**. Design aims to reduce these through channelization, one-way movements, or grade separation (flyovers).
- **Design Speed:** The speed of the approaching roads dictates the **turning radii**, the length of deceleration lanes, and the overall geometry of the intersection.
- **Sight Distance: Stopping Sight Distance (SSD) and Intersection Sight Distance (ISD)** are critical. Drivers must have an unobstructed view of the "sight triangle" to perceive and react to conflicting traffic.
- **Vulnerable Road Users (VRUs):** Design must include dedicated infrastructure for pedestrians and cyclists, such as refuge islands, tactile paving, and proper signal timing (pedestrian clearance intervals).
- **Geometric Characteristics:** This includes the number of legs, the angle of intersection (ideally **90 degrees**), and the provision of auxiliary lanes (left/right turn pockets) to prevent through-traffic delays.
- **Lighting and Drainage:** Adequate illumination is required to reduce night-time accidents. Proper longitudinal and cross-slopes (camber) must be maintained to prevent "aquaplaning" or water pooling at the junction.

10. Explain the factors that influence the selection of an airport location. How these factors impact the safety management systems and aerodrome certification process? Describe. 5+5=10

Answer:

I. Factors Influencing Airport Site Selection

The selection of an airport site is a complex multi-criteria decision-making process aimed at ensuring long-term viability and safety.

- **Meteorological Conditions:** Analysis of prevailing winds using a **Wind Rose Diagram** is critical to ensure at least 95% usability of runways. Persistent fog, crosswinds, and extreme temperatures (which affect air density and runway length requirements) are decisive factors.
- **Topography and Obstruction Limitation Surfaces (OLS):** The site must provide a clear approach and departure path. Steep terrain or man-made obstacles may violate **ICAO Annex 14** standards for OLS, limiting the types of aircraft that can operate.
- **Geotechnical and Hydrological Factors:** Soil bearing capacity (CBR value) influences pavement design and cost. Adequate drainage is essential to prevent hydroplaning and soil erosion.
- **Environmental & Land Use:** Proximity to bird sanctuaries or landfills (increasing bird strike risks) and noise contours affecting residential zones must be evaluated.
- **Proximity to Demand & Infrastructure:** The site must be close enough to urban centers to be economically viable but distant enough to allow for future expansion and integration with ground transport.

II. Impact on Safety Management Systems (SMS) and Aerodrome Certification

The physical characteristics of the chosen site dictate the complexity of the safety framework required for the aerodrome to be legal and functional.

- **Impact on SMS (Safety Management Systems):**
 - **Hazard Identification:** A site with unique geographic challenges (e.g., a "table-top" runway or high-altitude effects) requires a more robust **Hazard Log**.
 - **Risk Mitigation:** If a site is prone to specific weather (e.g., the "Katamari" wind in Nepal), the SMS must mandate specialized training and **Standard Operating Procedures (SOPs)** for low-visibility or high-crosswind landings.
 - **Bird Strike Management:** Sites near wetlands require an intensive Bird/Wildlife Hazard Management Plan as part of the SMS.
- **Impact on Aerodrome Certification (ICAO Annex 14 / CAAN Regulations):**
 - **Aerodrome Manual:** The manual must document how site-specific risks are managed. Any deviation from standard physical characteristics must be formally recorded.
 - **Aeronautical Studies:** When a site cannot meet standard "cleared zone" requirements (e.g., a hill obstructing an approach path), the certification process requires an **Aeronautical Study**. This study must prove that the risk is "As Low As Reasonably Practicable" (ALARP) through compensatory measures.
 - **Exemptions and Conditions:** If the site is non-compliant with standard safety distances, the National Aviation Authority (e.g., **CAAN**) may issue a conditional certificate, requiring stricter monitoring, higher lighting intensity, or specific pilot certification.

Section-D (20 Marks)

11. What are the factors affecting the self purification of natural streams? Explain. Discuss on the concept of oxygen Sag Curve in reference to self purification of natural streams. 10

Answer:

a. Factors Affecting Self-Purification

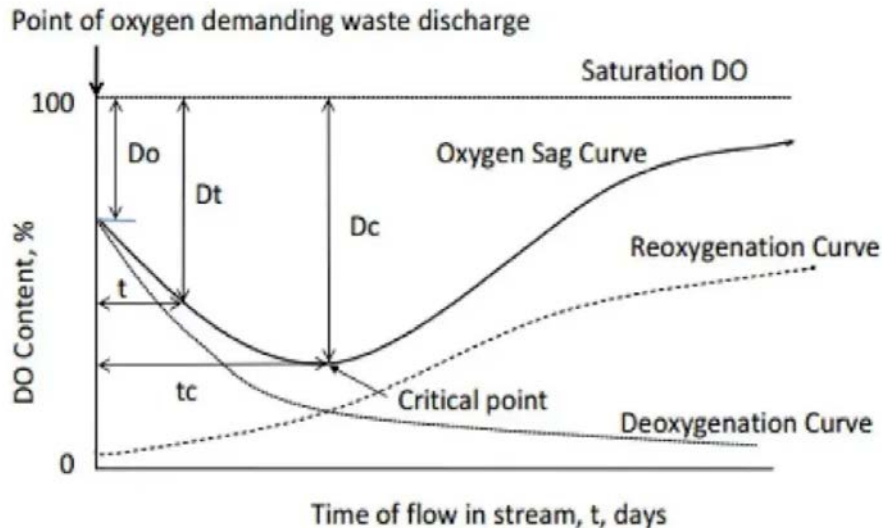
Self-purification is the natural process by which a body of water recovers its original quality after being polluted. The efficiency of this process depends on:

- **Dilution and Dispersion:** Large volumes of water reduce the concentration of pollutants. High velocity helps disperse organic matter, preventing localized high-concentration "slugs."
- **Current and Turbulence:** Turbulent flow increases the surface area contact with the atmosphere, significantly accelerating the **re-aeration rate**.
- **Temperature:**
 - *Positive effect:* Higher temperatures increase the metabolic rate of bacteria, leading to faster stabilization of organic matter.
 - *Negative effect:* The solubility of Dissolved Oxygen (DO) decreases as temperature rises, which can stress the aquatic ecosystem.
- **Sunlight:** Sunlight facilitates photosynthesis in algae and aquatic plants, which releases pure oxygen into the water during the day. It also acts as a natural disinfectant (UV radiation).

- **Sedimentation:** In slow-moving reaches, heavier organic solids settle to the bed (forming sludge banks), physically removing oxygen-demanding load from the main water column.
- **Oxidation and Reduction:** Bio-chemical oxidation converts unstable organic matter into stable inorganic compounds (CO_2, NO_3, SO_4)

b. **The Oxygen Sag Curve (Streeter-Phelps Concept)**

The Oxygen Sag Curve is a graphical representation of the relationship between the **De-oxygenation rate** and the **Re-aeration rate** in a stream. It plots the Dissolved Oxygen (DO) deficit against time or distance downstream from the point of pollution.



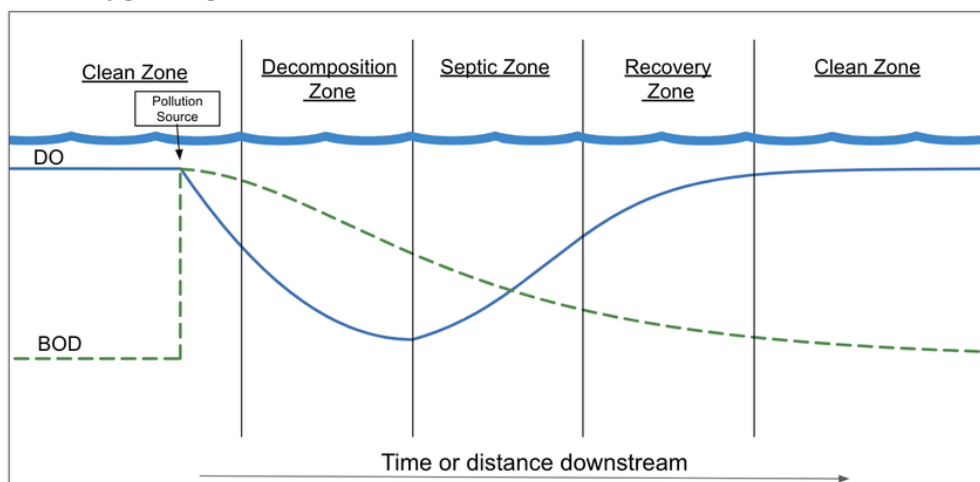
o **The Governing Mechanism**

The curve is the result of two competing simultaneous processes:

- **De-oxygenation (k_1):** The depletion of DO by aerobic bacteria as they decompose organic matter (BOD).
- **Re-aeration (k_2):** The replenishment of oxygen from the atmosphere through the water surface.
- **The Equation:** The deficit at any time t is given by the Streeter-Phelps equation:

$$D_t = \frac{k_1 L_0}{k_2 - k_1} [e^{-k_1 t} - e^{-k_2 t}] + D_0 e^{-k_2 t}$$

o **Phases of the Oxygen Sag Curve**



- **Zone of Degradation:** Located immediately below the sewage outfall. Water becomes turbid; DO begins to drop rapidly; fish life starts to move away.
- **Zone of Active Decomposition:** This is the most heavily polluted zone. DO may drop to zero (anaerobic conditions) if the pollution is heavy. This zone contains the **Critical Point**, where the DO deficit is maximum and the rate of de-oxygenation equals the rate of re-aeration.
- **Zone of Recovery:** The organic load has been largely stabilized. The rate of re-aeration exceeds the rate of de-oxygenation. DO levels begin to rise, and biodiversity starts to return.
- **Zone of Clear Water:** The stream reaches its original self-purified state. DO is back to saturation levels, and the water is chemically and biologically stable.

12. Answer the following.

- a) Estimate the stormwater volume generated during the heaviest rainfall in Godawari Municipality, Lalitpur using the Rational Method. The following information is given. 5

Population: 100,000, Rainfall: 240 mm in 24 hours, Area of the municipality: 70 square kilometers, Land cover: One fourth of the municipality is built-up, and the rest is covered by agriculture and forest. The formula is: $Q = CIA/360$ where; Q = Peak discharge (stormwater runoff) in cubic meters per second, C = Runoff coefficient, I = Rainfall intensity in mm/hour, A = Area in hectares.

- b) Estimate the sewage generated in the municipality assuming that per capita water consumption is 120 L per day and assuming evaporation loss of 20 percent. Compare the figure with the results obtained in (a). 5

Answer:

Step 1: Stormwater Calculation (Rational Method)

To calculate the peak discharge Q , we first determine the weighted runoff coefficient C based on land use and the rainfall intensity I .

- Area (A): $70 \text{ km}^2 = 7000 \text{ ha}$
- Land Use:
- Built-up: $\frac{1}{4} \times 7000 = 1750 \text{ ha}$ (Assume $C_1 = 0.8$)
- Agri/Forest: $\frac{3}{4} \times 7000 = 5250 \text{ ha}$ (Assume $C_2 = 0.3$)
- Weighted Runoff Coefficient (C):

$$C = \frac{(0.8 \times 1750) + (0.3 \times 5250)}{7000} = 0.425$$

- Rainfall Intensity (I):

$$I = \frac{240 \text{ mm}}{24 \text{ hr}} = 10 \text{ mm/hr}$$

- Peak Discharge (Q):

$$Q = \frac{CIA}{360} = \frac{0.425 \times 10 \times 7000}{360} \approx 82.64 \text{ m}^3/\text{sec}$$

Step 2: Sewage Generation Calculation

Sewage is calculated based on population water consumption, typically assuming 80% of water supplied returns as sewage, further reduced by evaporation.

- Total Water Consumption: $100,000 \times 120 \text{ L/day} = 12,000,000 \text{ L/day} = 12 \text{ MLD}$
- Initial Sewage Generation (80% of supply): $12 \times 0.8 = 9.6 \text{ MLD}$
- Net Sewage (after 20% evaporation loss):

$$\text{Net Sewage} = 9.6 \times (1 - 0.20) = 7.68 \text{ MLD}$$

- Conversion to m^3/sec :

$$7.68 \text{ MLD} = \frac{7680 \text{ m}^3}{24 \times 3600 \text{ sec}} \approx 0.089 \text{ m}^3/\text{sec}$$

The **Stormwater Runoff** ($82.64 \text{ m}^3/\text{sec}$) is vastly larger than the **Sewage Flow** ($0.089 \text{ m}^3/\text{sec}$).

Comparison and Significance:

The stormwater flow is approximately 928 **times** the volume of the daily sewage. In hilly municipalities like Godawari, this massive disparity necessitates **separate sewer systems**. A combined system would be economically unfeasible to build at such a scale and would result in the frequent overflow of untreated sewage into natural water bodies during rain events, as treatment plants cannot handle such extreme hydraulic surges.